

Rohit Goswami

Computational chemical kinetics, atomistic ML, rare events, and scientific software

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"An unproblematic state is a state without creative thought. Its other name is Death."

– David Deutsch

RESEARCH PROFILE

Researcher and scientific-software engineer working on Gaussian-process saddle point searches, machine-learned interatomic potentials, adaptive kinetic Monte Carlo, and reproducible HPC workflows for the long-time kinetics of activated processes.

Saddle search	GP-accelerated methods	JCTC, ChemPhysChem, ACS Phys. Chem. Au, and Frontiers papers on GP-accelerated NEB/dimer, adaptive pruning, benchmarking, and Hessian-aligned CI-NEB.
Atomistic ML	Interatomic potentials	EPFL labCOSMO work on metatensor/metatomic, tensor interoperability, GROMACS ML/MM, DLPack, and descriptor portability.
Long timescales	Adaptive KMC	Lead eOn maintainer; off-lattice/object KMC scaled past million-atom regimes at Queen's; absorbing-Markov superbasin escape methods.
Kinetics	Rate theory	PhD on efficient exploration of chemical kinetics; constrained-path, free-energy, and Bayesian uncertainty machinery for reaction rates.
Infrastructure	Reproducible science	NumPy/f2py maintainer, d-SEAMS co-creator, reproducible NEB workflows, pixi/Nix/Snakemake/CI practice, and JOSS editorial service.

TRACK RECORD

25 peer-reviewed and conference publications **1** Rannís grant delivered as PI **5+** years eOn maintenance **80+** JOSS reviews handled **575k** EUR COST Action co-proposer network

- Built and published a Gaussian-process saddle-search stack that reduces reference calls for difficult transition-state searches and now spans NEB, dimer, CI-NEB, adaptive pruning, and Bayesian benchmarking.
- Works at the interface between atomistic ML infrastructure and production simulation engines: `metatensor`, `metatomic`, GROMACS, LAMMPS, eOn, NumPy/f2py, and HPC packaging.
- Delivered an Icelandic Research Fund doctoral grant as principal investigator on Bayesian machine learning for magnetic interactions and saddle searches.

APPOINTMENTS

SURF B.V. 2026–present

HPC Advisor, Netherlands

Supports Dutch national research infrastructure and production HPC users across compute, workflow, and reproducibility.

Lab COSMO, EPFL 2025–2026

Systems Specialist, Switzerland

Developed `metatensor/metatomic` atomistic-ML infrastructure with Prof. Michele Ceriotti and Dr. Guillaume Fraux; worked on GROMACS ML/MM integration and zero-copy tensor exchange across C++, Python, Rust, and Fortran.

Queen's University 2025

Visiting Researcher, Canada

Scaled off-lattice/object kinetic Monte Carlo past million-atom regimes with Prof. Laurent Béland for long-time event-network evolution.

University of Iceland 2019–2024

Doctoral Researcher and Rannís Doctoral Grant PI, Reykjavík

Developed Gaussian-process accelerated saddle point search methods and Bayesian hierarchical benchmarks under Prof. Hannes Jónsson; ran the GP saddle-search work on Icelandic HPC infrastructure.

Quansight Labs 2021–2025

Software Engineer II, remote

Maintained `f2py` and contributed across NumPy/SciPy, meson-python, LFortran, OpenBLAS, PRIMA, and scientific build infrastructure.

EDUCATION

University of Iceland 2019–2025

Ph.D. Physical Sciences

Thesis: "Efficient Exploration of Chemical Kinetics." Supervisor: Prof. Hannes Jónsson; co-supervisor: Prof. Birgir Hrafnkelsson.

HBTU Kanpur 2014–2018

B.Tech. Chemical Engineering

First Division; computational nanoscience, ultrafast optics, and molecular-simulation analysis foundations.

PUBLICATIONS AND METHOD OUTPUTS

- Efficient Implementation of Gaussian Process Regression Accelerated Saddle Point Searches with Application to Molecular Reactions** JCTC, 2025
Core GP saddle-search method for molecular reactions, with true/reference force economy. [doi](#)
- A Tutorial Review of Bayesian Optimization with Gaussian Processes to Accelerate Stationary Point Searches** ACS Physical Chemistry Au, 2026
Review article framing GP acceleration as a reusable method for stationary-point search. [doi](#)
- Adaptive Pruning for Increased Robustness and Reduced Computational Overhead in Gaussian Process Accelerated Saddle Point Searches** ChemPhysChem, 2025
Cover-feature paper: adaptive training-set pruning for robust, cheaper saddle searches. [doi](#)
- Enhanced Climbing Image Nudged Elastic Band Method with Hessian Eigenmode Alignment** Frontiers in Chemistry, 2026
Hessian-aligned CI-NEB method for transition-state search, with Miha Gunde and Hannes Jónsson. [doi](#)
- Bayesian Hierarchical Models for Quantitative Estimates for Performance Metrics Applied to Saddle Search Algorithms** AIP Advances, 2025
Statistical benchmark layer for comparing saddle-search methods without over-reading small test sets. [doi](#)
- Reproducible Orchestration of Best Practices for Reaction Path Optimization with the Nudged Elastic Band** MethodsX, 2026
Executable workflow paper for reproducible NEB calculations and benchmarking. [doi](#)
- Two-Dimensional RMSD Projections for Reaction Path Visualization and Validation** MethodsX, 2026
Validation and inspection tool for reaction paths and transition-region geometry. [doi](#)
- Metatensor and Metatomic: Foundational Libraries for Interoperable Atomistic Machine Learning** JCP, 2026
Co-author on the tensor and model-stack infrastructure underpinning descriptor event memory. [doi](#)
- d-SEAMS: Deferred Structural Elucidation Analysis for Molecular Simulations** JCIM, 2020
Co-created molecular-simulation analysis engine for water, ice, and confined phases. [doi](#)
- The State of Fortran** Computing in Science & Engineering, 2022
Community paper and compiler/tooling context for maintainable scientific software. [doi](#)
- Flowy: High Performance Probabilistic Lava Emplacement Prediction** CPC, 2025
High-performance simulation workflow with Moritz Sallermann, Amrita Goswami, and Alejandro Peña-Torres. [doi](#)

Additional peer-reviewed and conference publications

- Study of Ice Nucleation on Silver Iodide Surface with Defects** Molecular Physics, 2019. [doi](#)
- High-Order Finite Element Method for Atomic Structure Calculations** Computer Physics Communications, 2023. [doi](#)
- Wailord: Parsers and Reproducibility for Quantum Chemistry** Proc. 21st Python in Science Conf. (SciPy), 2022. [doi](#)
- Reproducible High Performance Computing Without Redundancy with Nix** IEEE PDGC, 2023. [doi](#)
- High Throughput Reproducible Literate Phylogenetic Analysis** IEEE PDGC, 2023.
- Unified Software Design Patterns for Simulated Annealing** arXiv preprint, 2023. [doi](#)
- Compositional Analysis of Fragrance Accords Using Femtosecond Thermal Lens Spectroscopy** Chemistry – An Asian Journal, 2025. [doi](#)
- Translational and Rotational Dynamics of a Self-Propelled Janus Probe in Crowded Environments** Soft Matter, 2020. [doi](#)
- Diffusion of Self-Propelled Janus Tracer in Polymeric Environment** Preprint, 2017.
- Semi-Supervised Approaches to Ultrafast Pulse Shaping** ICOL-2019 (Springer), 2021. [doi](#)
- Qubit Network Barriers to Deep Learning** IEEE WRAP, 2019. [doi](#)
- Quantum Distributed Computing with Shaped Laser Pulses** OSA Photonics, 2016. [doi](#)
- Space Filling Curves: Heuristics for Semi-Classical Lasing Computations** URSI AP-RASC, 2019. [doi](#)
- Don't Pull Punches in Peer Review** Nature (Correspondence), 2019. [doi](#)

GRANTS, AWARDS, AND FUNDED ROLES

- COST Action CA24106 co-proposer and Iceland Management Committee member** 2025–2029
575,000 EUR European Cooperation in Science and Technology network: Building Education and One Health with Adaptive Convergence and Open Networks.
- Icelandic Research Fund Doctoral Research Grant, PI** 2020–2023
6.65M ISK. Title: “Magnetic interactions of itinerant electrons modeled using Bayesian machine learning.”
- Google Summer of Code mentor funding** 2023–2024
6,000 USD. Mentored Python bindings and contributor onboarding for d-SEAMS/pySEAMS.
- NumFOCUS Small Development Grants** 2022–2023
14,600 USD total. LFortran/SciPy and scientific Python compiler-build work.
- Google Season of Docs technical-writer grant** 2022
10,000 USD. Generated documentation capacity for scientific software.
- Springer Best Student Paper Award** 2016
Photonics conference recognition for early ultrafast-optics work.

OPEN SOURCE AND INFRASTRUCTURE RECORD

eOn: Lead maintainer for long-timescale dynamics: RPC architecture, GP saddle searches, Hessian CI-NEB, ML potentials, reproducible packaging.

metatensor/metatomic: Contributor to atomistic-ML infrastructure: DLPack, device-aware execution, Rust/C++/Python interoperability, GROMACS interface work.

f2py/NumPy: Maintainer of NumPy's Fortran-to-Python interface; Meson backend, derived-type support, and scientific build-system modernization.

d-SEAMS/pySEAMS: Co-creator of molecular-trajectory analysis engine; mentored Python bindings through Google Summer of Code.

SciPy, HiGHS, OpenBLAS, LFortran: Build, solver, compiler, and linear-algebra contributions used by the scientific Python stack.

Reproducibility stack: Nix, pixi, Snakemake, CMake, Meson, CI, containers, Slurm, Spack, EasyBuild, LaTeX, and literate org-mode workflows.

TALKS AND SCIENTIFIC COMMUNICATION

Managing concurrency for high throughput adaptive kinetic Monte Carlo

ACS Fall, 2025

Washington, DC. High-throughput aKMC concurrency and scheduling.

Towards a million atoms and a thousand KMC steps a day

Kernel aKMC, 2025

Queen's University, Canada. Scaling KMC/event-network methods with the Béland group.

Low Data Saddle Search Acceleration with Bayesian Performance Analysis

ML in Chemical and Materials Sciences, 2025

Talk on GP saddle-search acceleration and uncertainty-aware benchmarking.

Advocating for standardized APIs for common procedures

Psi-k, 2025

EPFL, Lausanne. Scientific-software interfaces for transition-state and atomistic workflows.

Accelerating Saddle Searches with Gaussian Process Regression

CECAM LTS-MAP, 2024

Invited talk, Lausanne. Long-time multi-scale simulations of activated events.

Computational Chemistry and Possible Tensor Applications

IPAM TMRC2, 2023

Invited talk, UCLA. Tensor methods and computational-chemistry workloads.

Development, Benchmarks and Workflows

CECAM-ESL Advancing the Molecular Paradigm, 2024

Reproducible benchmarking and workflow practice for electronic-structure codes.

TEACHING, MENTORSHIP, AND TRAINING

University of Iceland, supervisory and course teaching

2021–2024

Supervisory teacher for Software Quality Management; teaching in Machine Learning, Statistical Inference for Biology, and computational workshops.

Software/Data Carpentry and Code in Place

2020–2024

Instructor, helper, mentor, and section leader across Git/GitHub, R, Python, and introductory computing.

Google Summer of Code and NumFOCUS mentorship

2023–present

Mentored first-time scientific-software contributors; coordinated contributor pathways around NumFOCUS projects.

SELECTED SERVICE AND REVIEWING

- **Journal of Open Source Software editor** (2024–present), topic editor; handled 80+ reviews across scientific software.
- **Nordic Research Software Engineers Association**, board member and founding secretary.
- **IEEE P3173 vice chair** and IEEE standards work on chemical hazard labelling; earlier standards participation in P2881 and P1940.
- **NumFOCUS Google Summer of Code administrator** (2024–present).
- **National HPC Centre of Iceland consultant** (2021–2024), Simulation and Data Lab Computational Chemistry.
- **Triangle Scholarly Communication Institute** (2019), invited participant in a funded incubator on equity and diversity in peer review.
- Reviewer for Nature Communications, Physical Review Letters, Physical Review A/E/Applied, IOP journals, PLOS ONE, PeerJ, JOSS, and Manning Publications.

HONOURS AND FELLOWSHIPS

Fellowships and awards: IPAM Fellow, Institute for Pure and Applied Mathematics, UCLA (2021), Tensor Methods long programme; Springer Best Student Paper Award, Photonics-2016 (Nonlinear-Optics session, IIT Kanpur).

TECHNICAL BREADTH

Scientific domains: Chemical kinetics, transition-state theory, adaptive kinetic Monte Carlo, molecular dynamics, photocatalysis, atomistic machine learning, Bayesian optimization, Gaussian processes, uncertainty quantification, rare-event sampling.

Simulation, HPC, and languages: eOn, LAMMPS, GROMACS, MPI/OpenMP, CUDA, Slurm, Nix, Spack, EasyBuild, containers; C++, Python, Fortran, Rust, R, Julia, shell, with bindings/ABI work across the Python/C++/Fortran/Rust stacks.

Data and ML: PyTorch, scikit-learn, GPy, Stan, Bayesian hierarchical models, descriptor tensors, reaction-path validation, reproducible benchmark design.

Solvers and tooling: HiGHS LP/MIP integration into SciPy (linprog/milp) and PRIMA; OpenBLAS Meson port (default NumPy/SciPy BLAS/LAPACK); ASV benchmarking co-maintenance; wailord ORCA parser and rOpenSci-reviewed fastMatMR R bindings.

AFFILIATIONS

Institute of Physics (MInstP) • Royal Society of Chemistry (MRSC) • BCS, the Chartered Institute for IT (MBCS) • ACM • APS • IEEE • AIChE • Nordic-RSE • TeX Users Group.